Lawrence Lim

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Education

University of California, Santa Barbara — Santa Barbara, CA
Doctor of Philosophy (Ph.D.) in Computer Science — Class of 2026

University of California, Santa Barbara — Santa Barbara, CA
Bachelor of Science (B.S.) in Computer Science (3.94 GPA) — Class of 2020
College of Engineering Honors

Work Experience

Secure LLM Inference Researcher, Distributed Systems Lab
University of California, Santa Barbara — September 2020 – Current

- Implemented a 6x lower latency private LLM inference system on GPU with homomorphic encryption by developing novel algorithms, leading to an accepted paper at a top-tier conference [1] [3].
- Developed a 100x more efficient private token sampling algorithm for privacy-preserving LLM systems, resulting in an accepted paper at a premier AI conference [2].
- Optimized key rotations by developing new protocols to eliminate race conditions while utilizing new updatable encryption primitives, resulting in a research paper at a top-tier conference [6].
- Built augmented and virtual reality applications for spatialized music, resulting in accepted papers at top HCI conferences [7] [8].
- Led research teams, forged industry collaborations, mentored undergraduate and graduate students, gave talks at conferences and in industry, and secured $200K+ in grant funding.

Software Engineer Intern, Google
Sunnyvale, CA — June 2024 - September 2024

- Built a compiler for privacy-preserving AI that translates Python ML models into efficient CKKS homomorphic encryption programs, achieving a 3000x speedup over prior infrastructure.
- Implemented optimized matrix multiplication FHE kernels, leading to a 6x performance improvement.
- Developed a pipeline in MLIR, which directly contributed to the HEIR research paper [4].

Software Engineer Intern, Oracle
Redwood City, CA — June 2021 – September 2021, June 2022 - September 2022

- Developed algorithms for optimizing for net benefit of query plan native compilation by modeling native compilation benefit and clustering to group similar query plans, leading to a patent [5].

Software Engineer Intern, Toyon Research Corporation
Goleta, CA — July 2019 – September 2020

- Improved accuracy of object detection in applications of AI models like RetinaNet and Faster R-CNN by tuning hyperparameters and implementing image transformations.
- Obtained secret security clearance, authoring progress reports for government contracts.
- Created image classification model using ResNet to determine benefits of additional LIDAR data.

Skills

Artificial Intelligence:	Large Language Models, Transformers, CNNs
Security and Privacy:	Homomorphic Encryption, Multi-Party Computation, Differential Privacy
Systems:	Databases, Distributed Systems, Compilers, Spatial Computing

Selected Publications

[1] **Lim, L.**, Kalagi, V., Novick, J., Liu, J., Agrawal, D., and El Abbadi, A. "Tricycle: Private Transformer Inference with Tricyclic Encodings" (to appear at CCS 2026)
[2] **Lim, L.**, Liu, J., Kalagi, V., Agrawal, D., and El Abbadi, A. "Hyperion: Private Token Sampling with Homomorphic Encryption" (ACL 2026)
[3] **Lim, L.**, Agrawal, D., and El Abbadi, A. "Private LLM Inference with Homomorphic Encryption" (EDBT 2026, VLDB 2026)
[4] Ali, A., Choi, J., Gipson, B., Gorantala, S., Kun, J., Legiest, W., **Lim, L.**, Viand, A., Demissie, M., and Zheng, H. "HEIR: A Universal Compiler for Homomorphic Encryption" (2025)
[5] **Lim, L.**, Aghababae, L., and Swart, G. "Compiling Just the Best Bits: Finding the Maximum Benefit Shared Subexpressions in a Query Workload" (US patent 2025)
[6] **Lim, L.**, Goh, W., Agrawal, D., El Abbadi, A., Gupta, T. "QUICKeR: Quicker Updates Involving Continuous Key Rotation" (PoPETS 2024)
[7] Jiang, S., **Lim, L.**, and Sra, M. "Spatializing Music in Virtual Reality" (SUI 2023)
[8] **Lim, L.**, Goh, W., and Sra, M. "A Spatial Music Listening Experience in Augmented Reality" (UIST 2021)